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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-10 & MVD-11 - 20kV Centralized MVD Room Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																												
D									D																																												
E									E																																												
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="3"> ABB EPMV PT. ABB Sakti Industri </td><td rowspan="3">COVER SHEET SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on:</td><td>BE318042</td><td></td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 28</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV PT. ABB Sakti Industri	COVER SHEET SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont. 28	F
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	==MVD-10_MVD-11=H00&GAD/2		INDEX OF SHEETS		0	1										
B	==MVD-10_MVD-11=H00&GAD/3		GENERAL CHARACTERISTICS		0	1										
	==MVD-10_MVD-11=H00&GAD/4		REFERENCE DESIGNATIONS		0	1										
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	C	==MVD-10_MVD-11=H00&GAD/9		SINGLE LINE DIAGRAM		0	1									
==MVD-10_MVD-11=H00&GAD/10		SINGLE LINE DIAGRAM		0	1											
==MVD-10_MVD-11=H00&GAD/11		SINGLE LINE DIAGRAM		0	1											
==MVD-10_MVD-11=H00&GAD/12		SINGLE LINE DIAGRAM		0	1											
==MVD-10_MVD-11=H00&GAD/13		INSTALLATION RULES		0	1											
==MVD-10_MVD-11=H00&GAD/14		FRONT VIEW & GA		0	1											
==MVD-10_MVD-11=H00&GAD/15		FRONT VIEW & GA		0	1											
==MVD-10_MVD-11=H00&GAD/16		FRONT VIEW & GA		0	1											
==MVD-10_MVD-11=H00&GAD/17		FOUNDATION FRAMES		0	1											
D		==MVD-10_MVD-11=H00&GAD/18		FOUNDATION FRAMES		0	1									
	==MVD-10_MVD-11=H00&GAD/19		FOUNDATION FRAMES		0	1										
	==MVD-10_MVD-11=H00&GAD/20		FOUNDATION FRAMES DETAILS		0	1										
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	==MVD-10_MVD-11=H00&GAD/23		SECTION VIEW		0	1										
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	==MVD-10_MVD-11=H00&GAD/27		SECTION VIEW		0	1										
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E									E							
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	INDEX OF SHEETS		Scale		= H00	F
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A						
B	SWITCHGEAR VERSION = UNISEC				COLOUR OF FRONT DOORS = RAL 7035				B						
	RATED VOLTAGE (Un) = 24 kV				SIDEWALLS PAINTED = YES										
	OPERATING VOLTAGE = 20 kV				SIDES = L+R										
	RATED FREQUENCY (fr) = 50 Hz														
C	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up) = 125 kV				SUPPLIED IMPLEMENTS				C						
	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud) = 50 kV														
	RATED CURRENT OF MAIN BUSBARS (Ir) = 1250 A														
	RATED SHORT-TIME WITHSTAND CURRENT (Ik) = 20 kA														
D	RATED PEAK WITHSTAND CURRENT (Ip) = 50 kA				IDENTIFICATION OF CONDUCTORS				D						
	RATED DURATION OF SHORT CIRCUIT (tk) = 1 s														
	INTERNAL ARC CLASSIFICATION (IAC) = AFLR / 16kAx1s														
	AMBIENT CONDITION = NORMAL														
E	AMBIENT AIR TEMPERATURE = -5°C...+40°C				FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW -XDA 9 10 11 0T:VDX- -XDA:10 -XDA:10				E						
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED) = IP3X														
	DEGREE OF PROTECTION WITH OPEN DOORS = IP2X														
	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua) = 230VAC														
F	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua) = 230VAC				Diagram showing conductor identification: -XDA connected to terminal 10 of component -XA:10.				F						
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua) = 230VAC														
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = 450/750V														
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS = PVC														
F	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS = 2.5 mm²/BLACK				Legend for conductor identification: 0T:VDX- -XDA:10 9 10 11 -XDA:10										
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS = 1.5 mm²/BLACK														
	CROSS-SECTION OF CONDUCTORS FOR MOTORS = 2.5 mm²/BLACK														
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS) = 1.5 mm²/BLACK														
F	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS = 1.5 mm²/BLACK				Legend for conductor identification: 0T:VDX- -XDA:10 9 10 11 -XDA:10										
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE = 2.5 mm²/BLACK														
	COMMUNICATION CABLE = N/A														
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION	-CA	CAPACITORS
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION	-CA1	CAPACITOR FOR FRONT VENTILATOR
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)	-CA2	CAPACITOR FOR REAR VENTILATOR
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION	-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION	-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT
B	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION	-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION	-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION	-EB1	HEATER LOCATED IN CABLE COMPARTMENT
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION	-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT
	-BGK	POSITION SWITCH OF KEY LOCK	-EB3	HEATER LOCATED IN MOTOR CABINET
C	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1	-EB4	HEATER FOR VENTILATION TEST
	-BGS1....-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS	-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION	-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION	-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION	-FA1	SURGE ARRESTER LOCATED ON PHASE L1
D	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION	-FA2	SURGE ARRESTER LOCATED ON PHASE L2
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION	-FA3	SURGE ARRESTER LOCATED ON PHASE L3
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION	-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION	-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION	-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2
E	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION	-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION	-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION	-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR	-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1
	-BGV2	POSITION SWITCH OF REAR VENTILATOR	-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2
F	-BM	HYGROSTAT	-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-FCT	THERMAL OVERLOAD RELEASES
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GA	GENERATORS
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GQ1	FRONT VENTILATOR
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GQ2	REAR VENTILATOR
G	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER	-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2	-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3	-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE
	-BR	FLAME DETECTORS, SMOKE DETECTORS	-KFA3....-KFA9	AUXILIARY RELAYS OR CONTACTORS
	-BT	THERMOSTAT	-KFC	CLOSING RELAYS OR CONTACTORS
H	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS	-KFI	INTEGRATED CIRCUITS
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS	-KFL	LOCKOUT RELAY
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM	-KFM	MICROPROCESSORS
	-BU	BUCHHOLZ RELAY	-KFN	ANTIPUMPING RELAYS
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING	-KFO	OPENING RELAYS OR CONTACTORS
I	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING	-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)
	-BZ,-BZ4	MULTIFUNCTION SENSORS	-KFR	RECLOSING RELAYS
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1	-KFS	SYNCHRONIZING DEVICES
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2	-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3	-KFZ	CONTROLLERS
J			-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER
			-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD
			-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1
			-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2
			-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS

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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant				SLD & GA Drawing		Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri		MVD-10 & MVD-11 - 20kV Centralized MVD Room		Drawing Nbr.		Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042						Customer Doc. Nbr.		Cont. 28

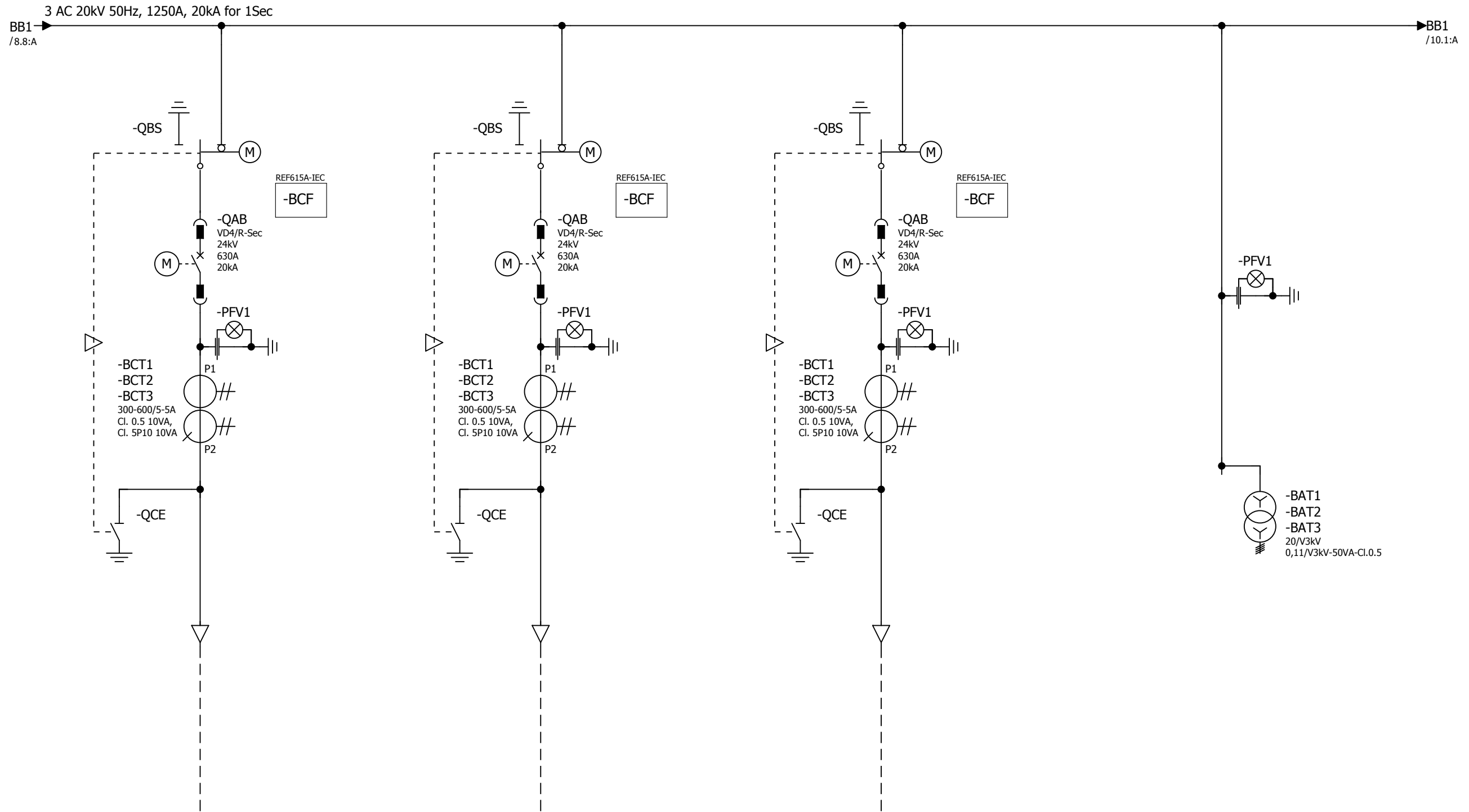
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A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A							
	-TFC		CURRENT TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING									
	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION									
	-TFJ		ACTIVE ENERGY TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS									
	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS									
	-TFM		MULTIFUNCTION TRANSDUCERS			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS									
	-TFP		POWER-FACTOR TRANSDUCERS		-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS									
	-TFQ		REACTIVE POWER TRANSDUCERS			TERMINAL BLOCK FOR SPECIAL USE									
B	-TFS		SIGNAL CONVERTERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD		B							
	-TFV		VOLTAGE TRANSDUCERS			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS									
	-WA		BUSBARS		-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT									
	-WBC		POWER CABLES			CURRENT SOCKETS									
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDS	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB									
	-WGC		CONTROL CABLES			CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK									
	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK									
	-WH		OPTICAL FIBRES			TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS									
	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1									
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2									
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2									
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3									
	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3									
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR									
	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR									
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			SUPPORT TERMINAL BLOCK									
	-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE									
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE									
	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE									
	-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE									
	-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE									
	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE									
D	-XDC		CUSTOMER TERMINAL BLOCK		-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE									
	-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE									
	-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE									
	-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE									
	-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS		-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE									
	-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT									
	-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE		-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT									
	-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE												
	-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS												
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR												
	-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS												
	-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS												
	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)												
	-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)												
F									F						
1		2		3		4		5		6		7		8	

			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SINGLE LINE DIAGRAM SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 8	
Revision	Date	Name	Approved	AI		Based on: BE318042	Customer Doc. Nbr.				Cont. 28

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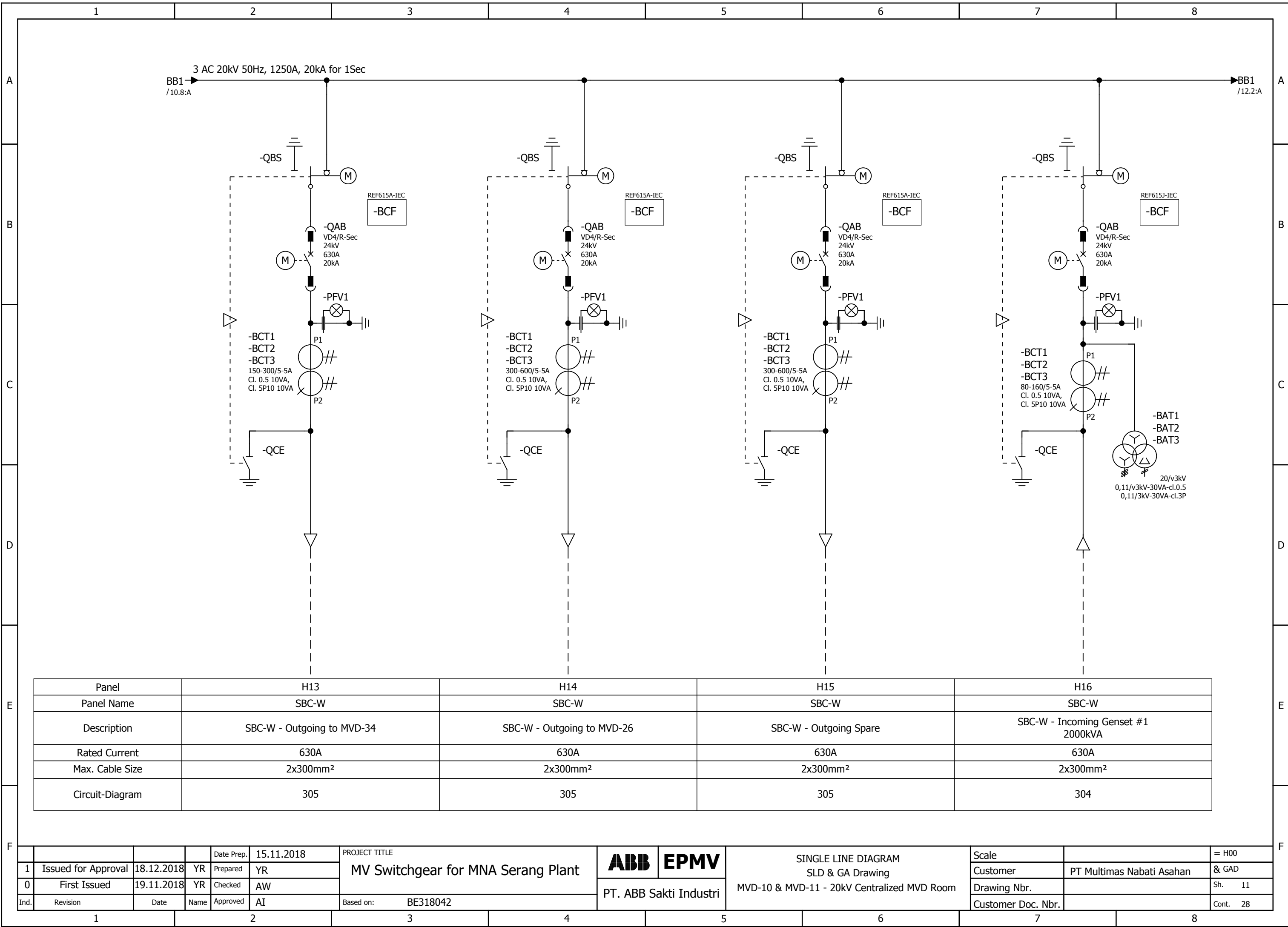
Panel	H05	H06	H07	H08
Panel Name	SBC-W	SBC-W	SBC-W	DRC
Description	SBC-W - Outgoing to MVD-33	SBC-W - Outgoing to MVD-25	SBC-W - Outgoing Spare	DRC - Bus PT
Rated Current	630A	630A	630A	-
Max. Cable Size	2x300mm ²	2x300mm ²	2x300mm ²	-
Circuit-Diagram	305	305	305	308

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SINGLE LINE DIAGRAM SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 9		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

Panel	H09	H10	H11	H12
Panel Name	WBS	DRS	SBC-W	SBC-W
Description	WBS - Bus Tie	DRS - Riser with VT	SBC-W - Outgoing to MVD-21	SBC-W - Outgoing to MVD-30
Rated Current	1250A	1250A	630A	630A
Max. Cable Size	-	-	2x300mm ²	2x300mm ²
Circuit-Diagram	303	309	305	305

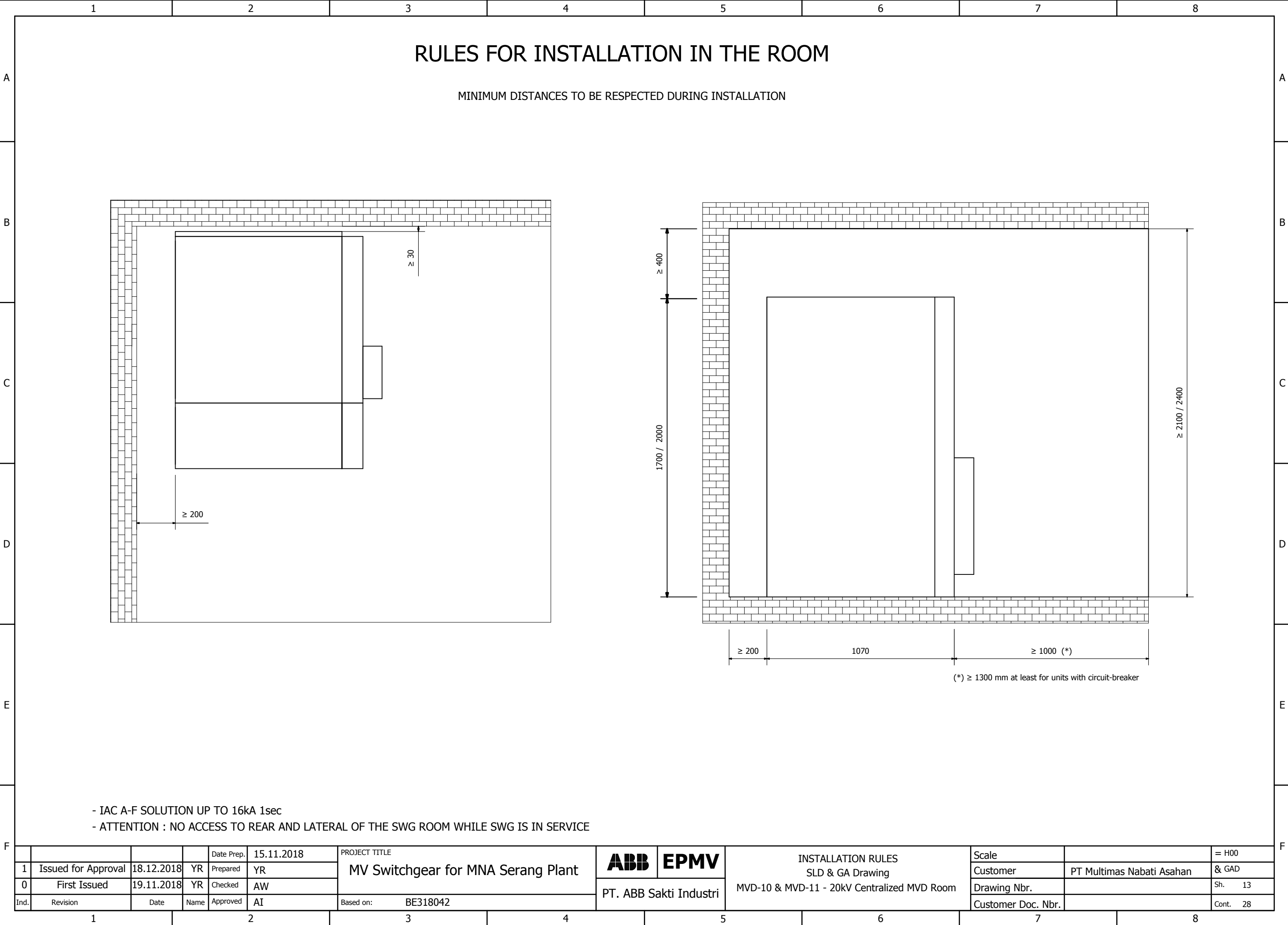
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SINGLE LINE DIAGRAM SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 10	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 28	

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			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SINGLE LINE DIAGRAM SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 12	
Revision	Date	Name	Approved	AI		Based on: BE318042	Customer Doc. Nbr.				Cont. 28

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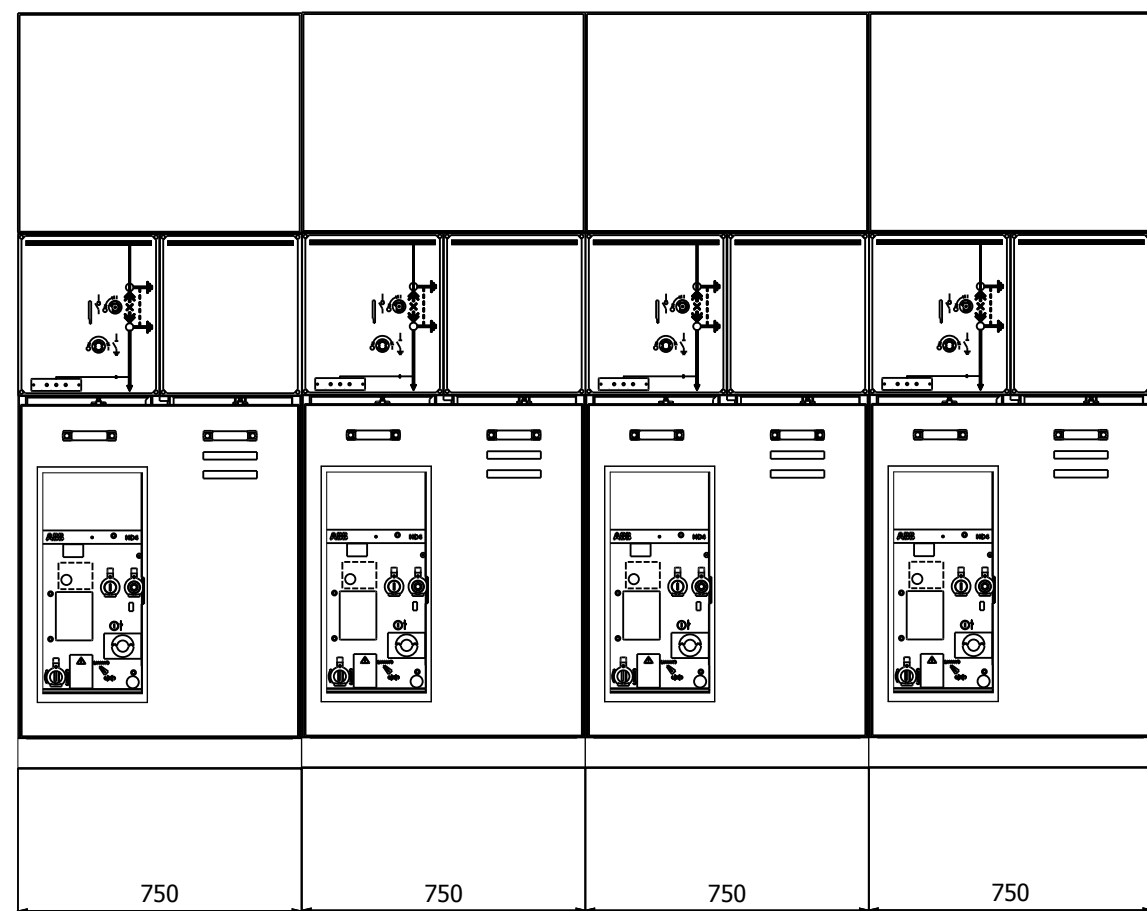
Panel	H01	H02	H03	H04	H05	H06	H07	H08
Panel Type	WBC	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W	DRC
Description	WBC - Incoming	SBC-W - Outgoing	SBC-W - Spare Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 14	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 28		

Panel	H09	H10	H11	H12	H13	H14	H15	H16
Panel Type	WBS	DRS	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W
Description	WBS - Bus Tie	DRS - Riser	SBC-W - Spare Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Incoming Genset

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 15	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 28		

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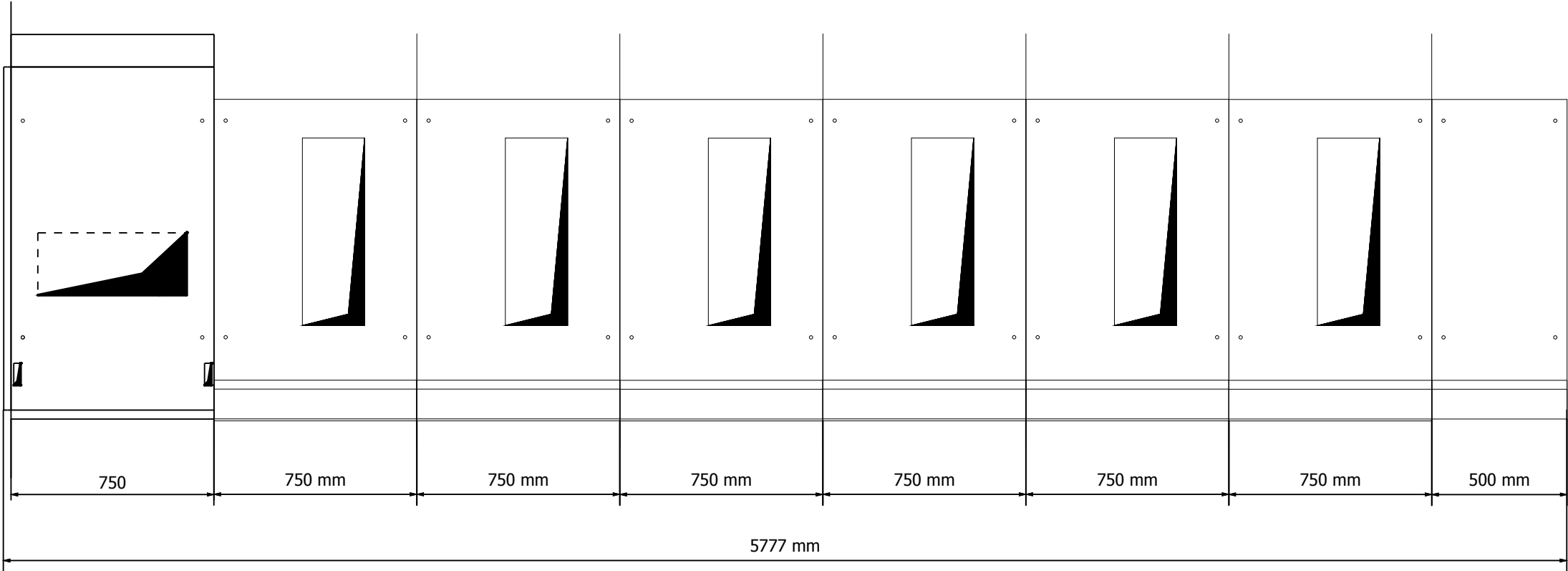


Panel	H17	H18	H19	H20
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W
Description	SBC-W - Incoming Genset	SBC-W - Incoming Genset	SBC-W - Incoming Genset	SBC-W - Incoming Genset

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 16	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.				Cont. 28

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FOUNDATION FRAMES

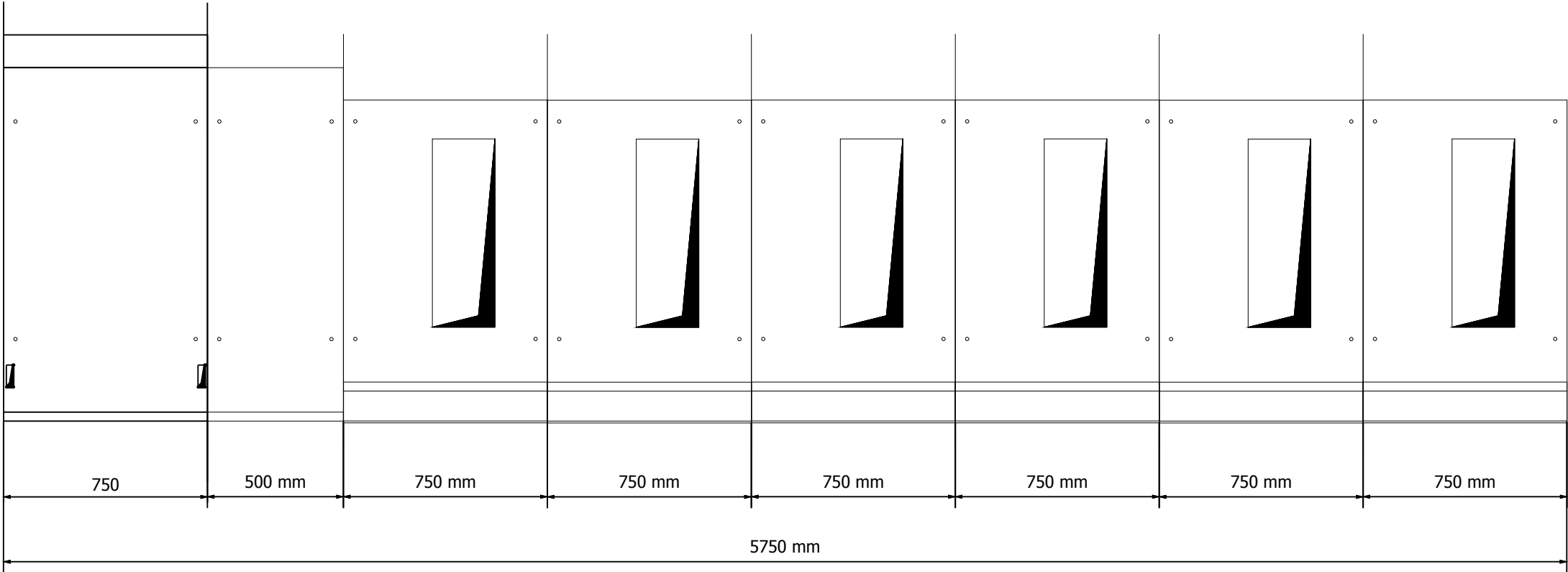


Panel	H01	H02	H03	H04	H05	H06	H07	H05
Panel Type	WBC	SBC-W	SBC-W	SBC-W	WBC	SBC-W	SBC-W	DRC
Description	WBC - Incoming	SBC-W - Outgoing	SBC-W - Spare Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR						Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW						Drawing Nbr.		Sh. 17
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

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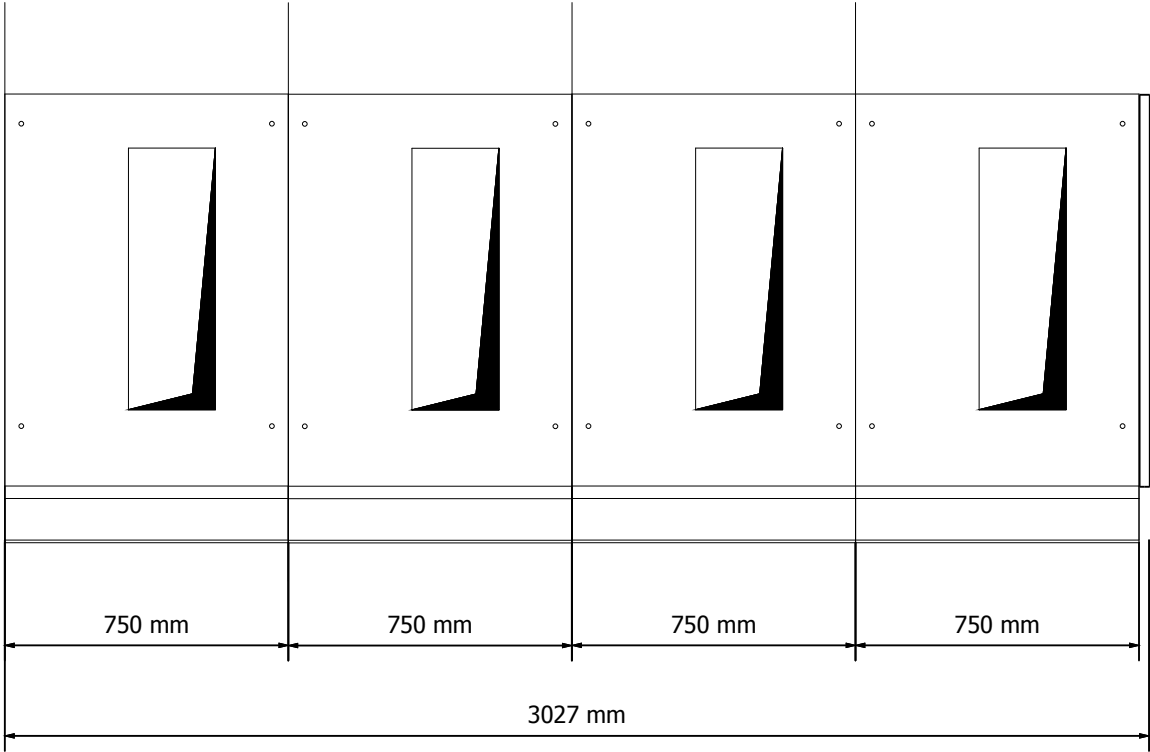


Panel	H09	H10	H11	H12	H13	H14	H15	H16
Panel Type	WBS	DRS	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W
Description	WBS - Bus Tie	DRS - Riser	SBC-W - Spare Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Outgoing	SBC-W - Incoming Genset

Ind.				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room		Scale		= H00
	1	Issued for Approval	18.12.2018	YR	Prepared						Customer	PT Multimas Nabati Asahan	& GAD
	0	First Issued	19.11.2018	YR	Checked						Drawing Nbr.		Sh. 18
	Revision		Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 28

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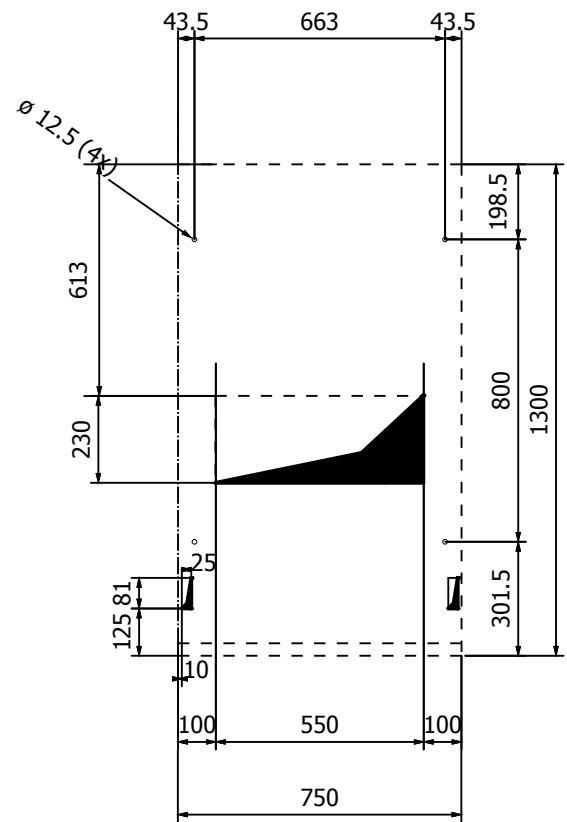
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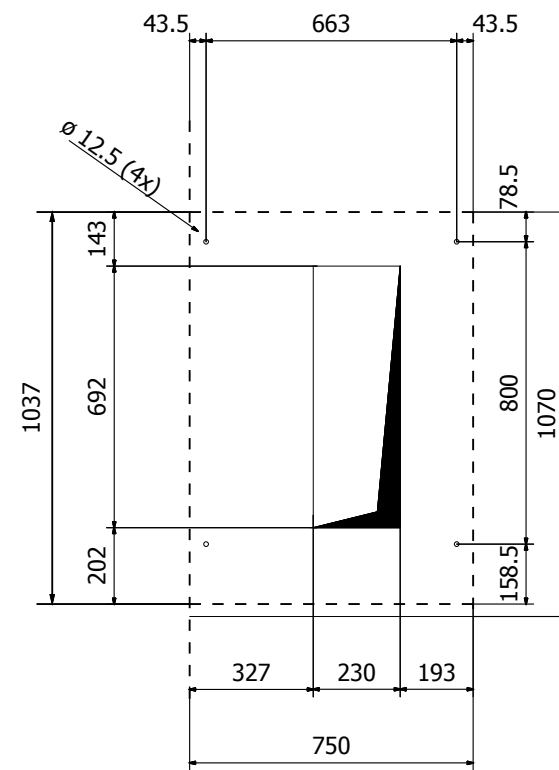
Panel	H17	H18	H19	H20
Panel Type	SBC-W	SBC-W	SBC-W	SBC-W
Description	SBC-W - Incoming Genset	SBC-W - Incoming Genset	SBC-W - Incoming Genset	SBC-W - Incoming Genset

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri			Drawing Nbr.		Sh. 19
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042			Customer Doc. Nbr.		Cont. 28

FOUNDATION FRAMES DETAILS



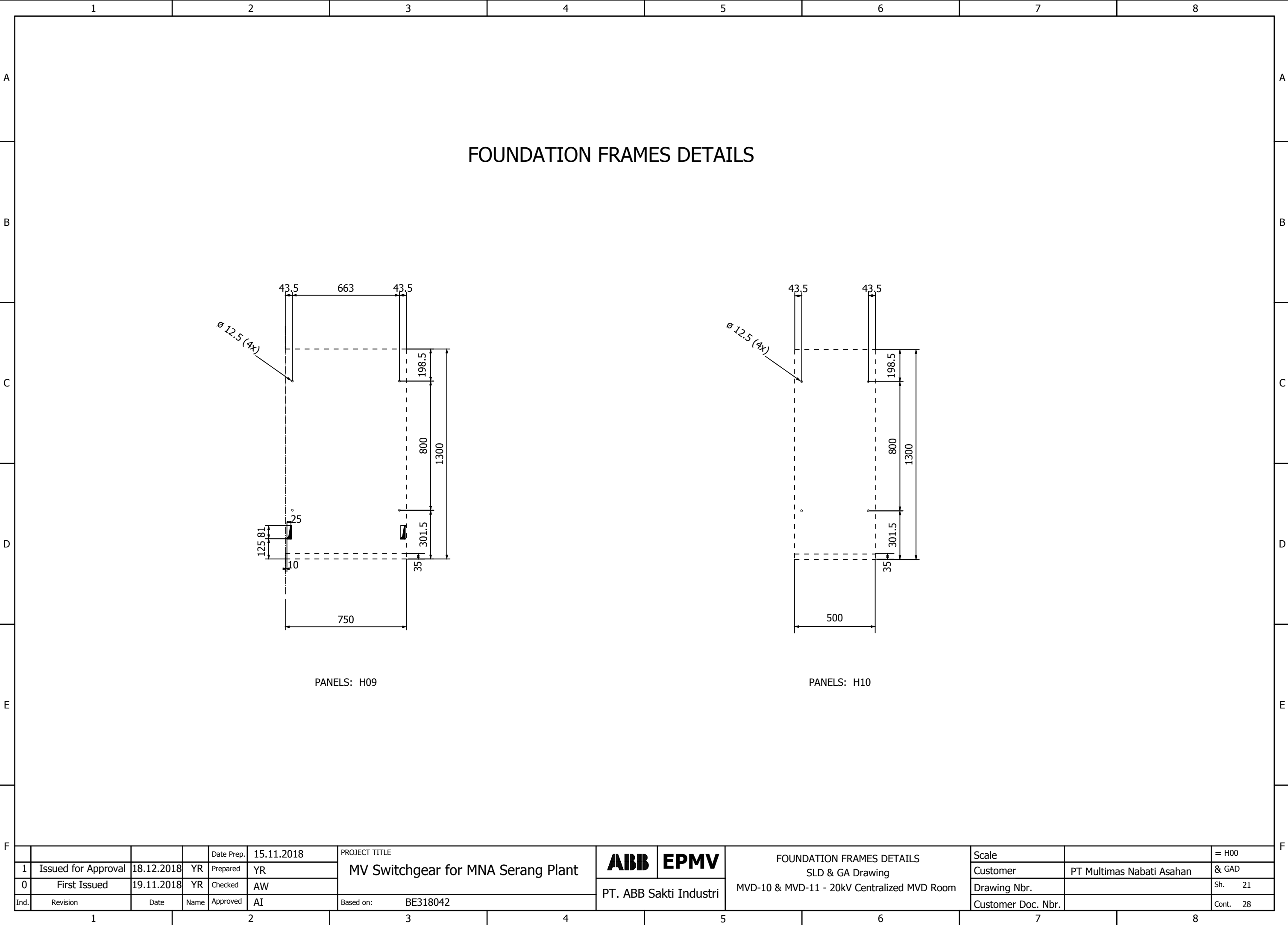
PANELS: H01



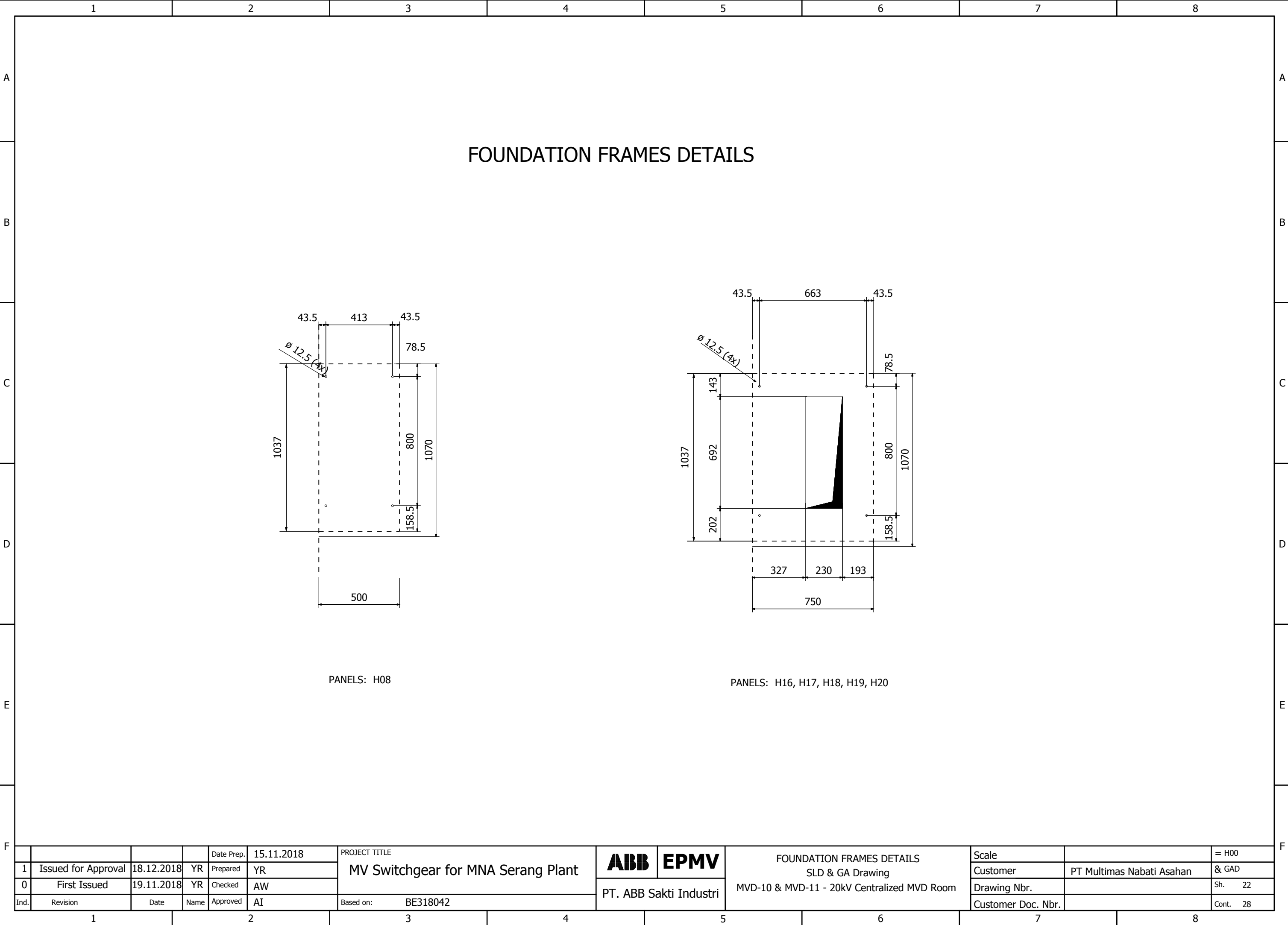
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES DETAILS SLD & GA Drawing	MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00	
1	Issued for Approval	18.12.2018	YR	Prepared	YR						PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW							Drawing Nbr.		Sh. 20
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.		Cont. 28					

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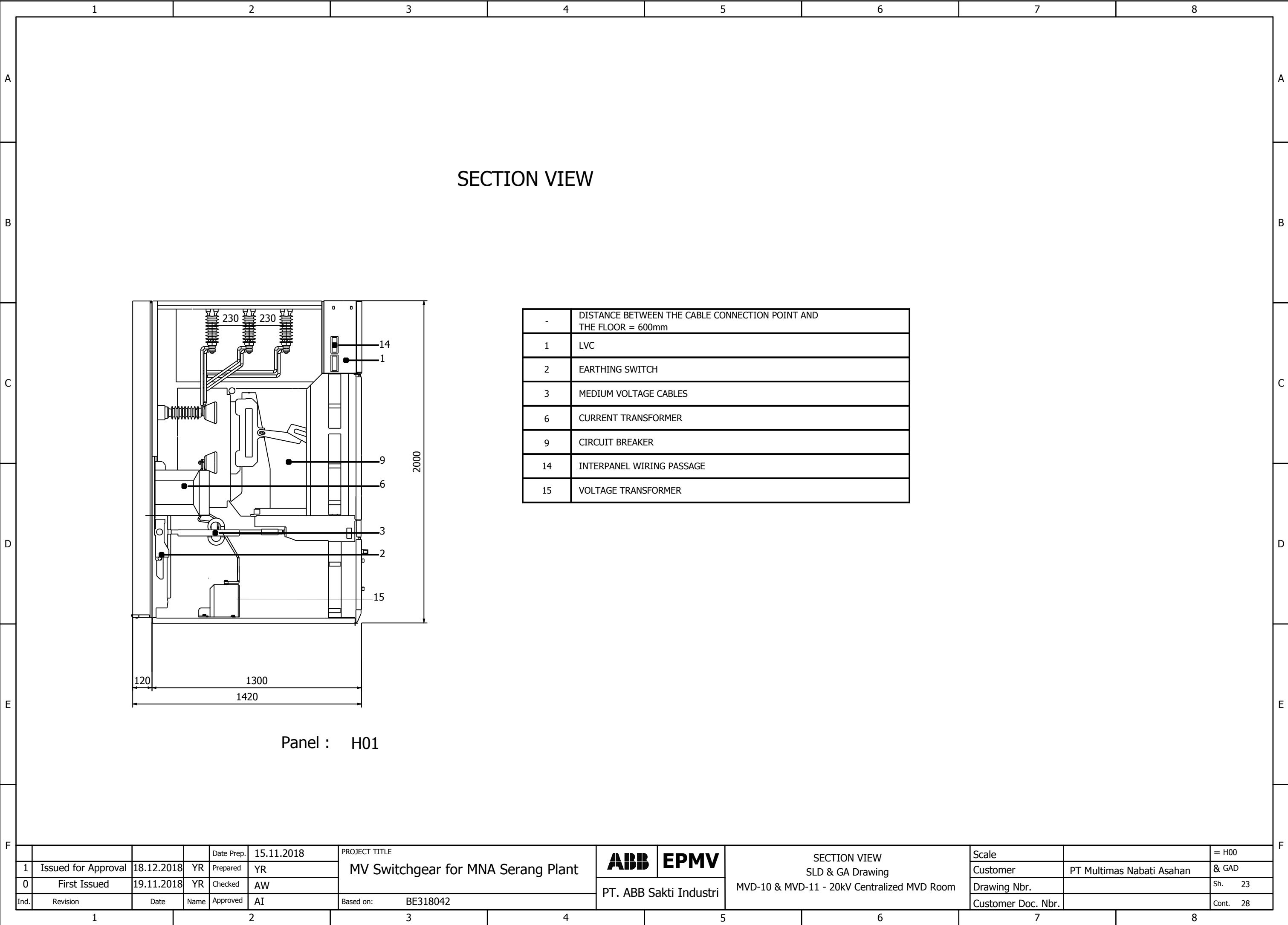


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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV PT. ABB Sakti Industri	FOUNDATION FRAMES DETAILS SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW					Drawing Nbr.		Sh. 22
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042			Customer Doc. Nbr.		Cont. 28

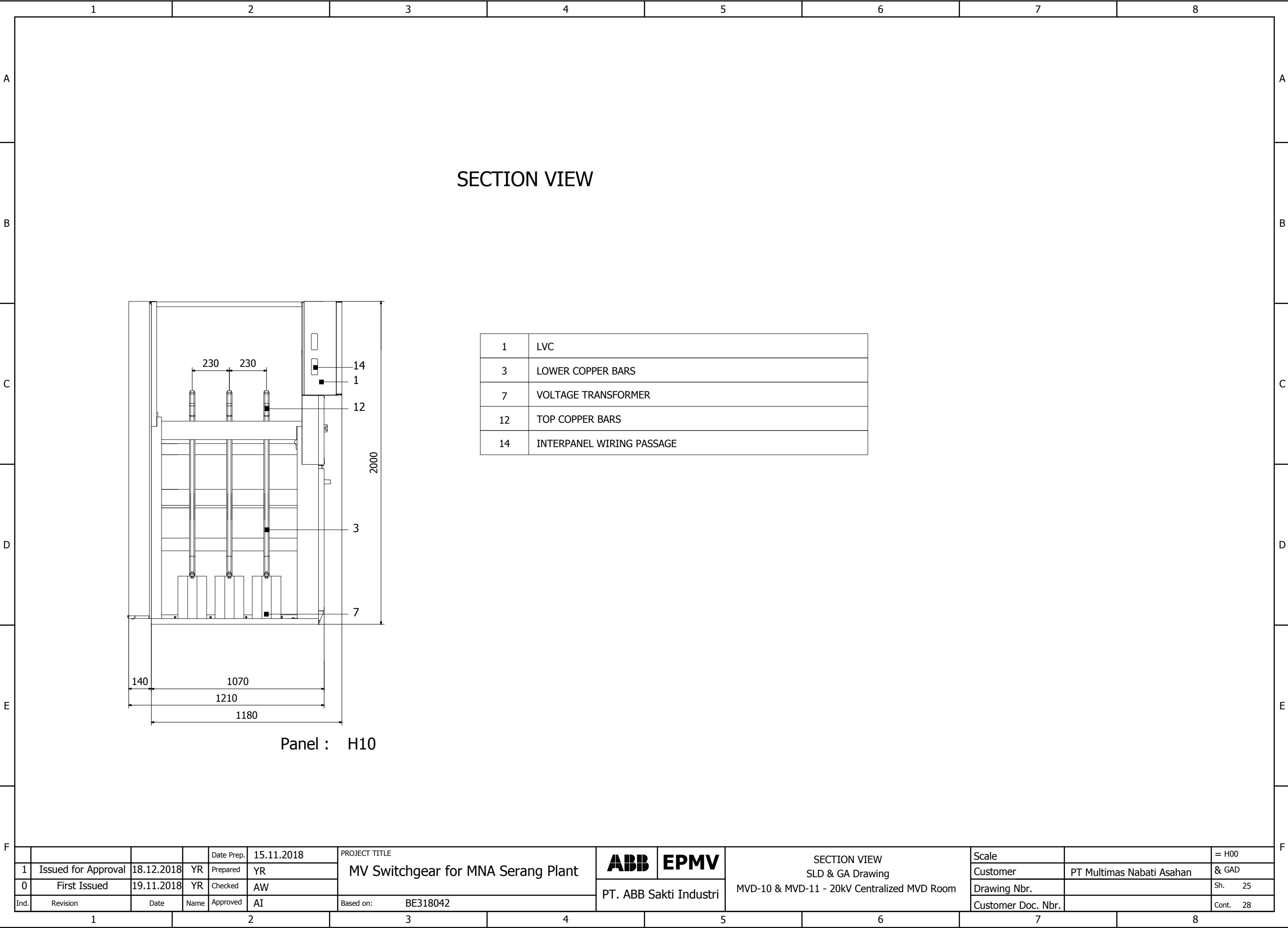
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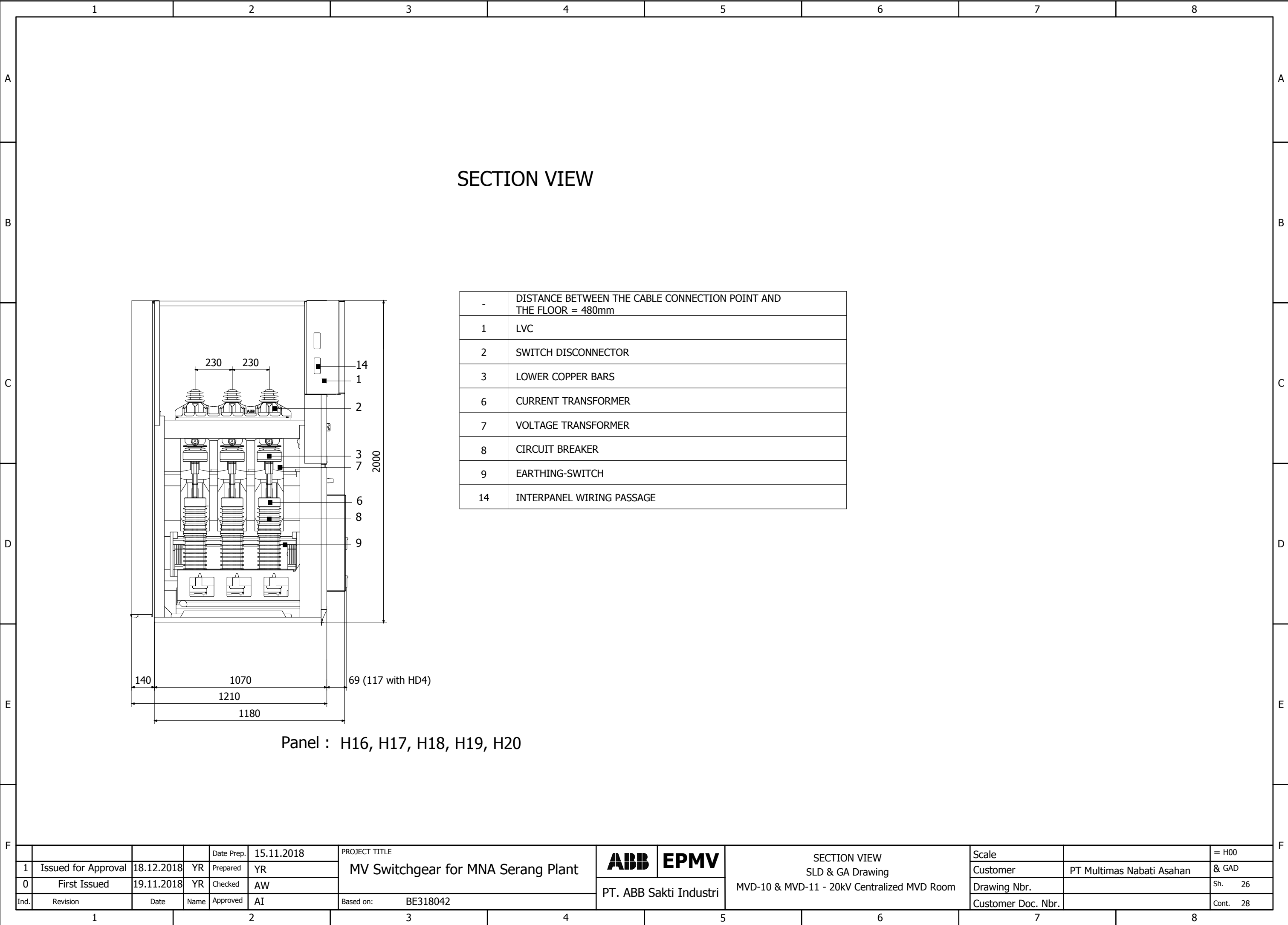
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1		2		3		4		5		6		7		8	
SECTION VIEW															
Panel : H09															
				Date Prep.	15.11.2018	PROJECT TITLE			ABB	EPMV	SECTION VIEW		Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					SLD & GA Drawing		Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW				PT. ABB Sakti Industri	MVD-10 & MVD-11 - 20kV Centralized MVD Room	Drawing Nbr.		Sh.	24	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042					Customer Doc. Nbr.		Cont.	28	
1		2		3		4		5		6		7		8	

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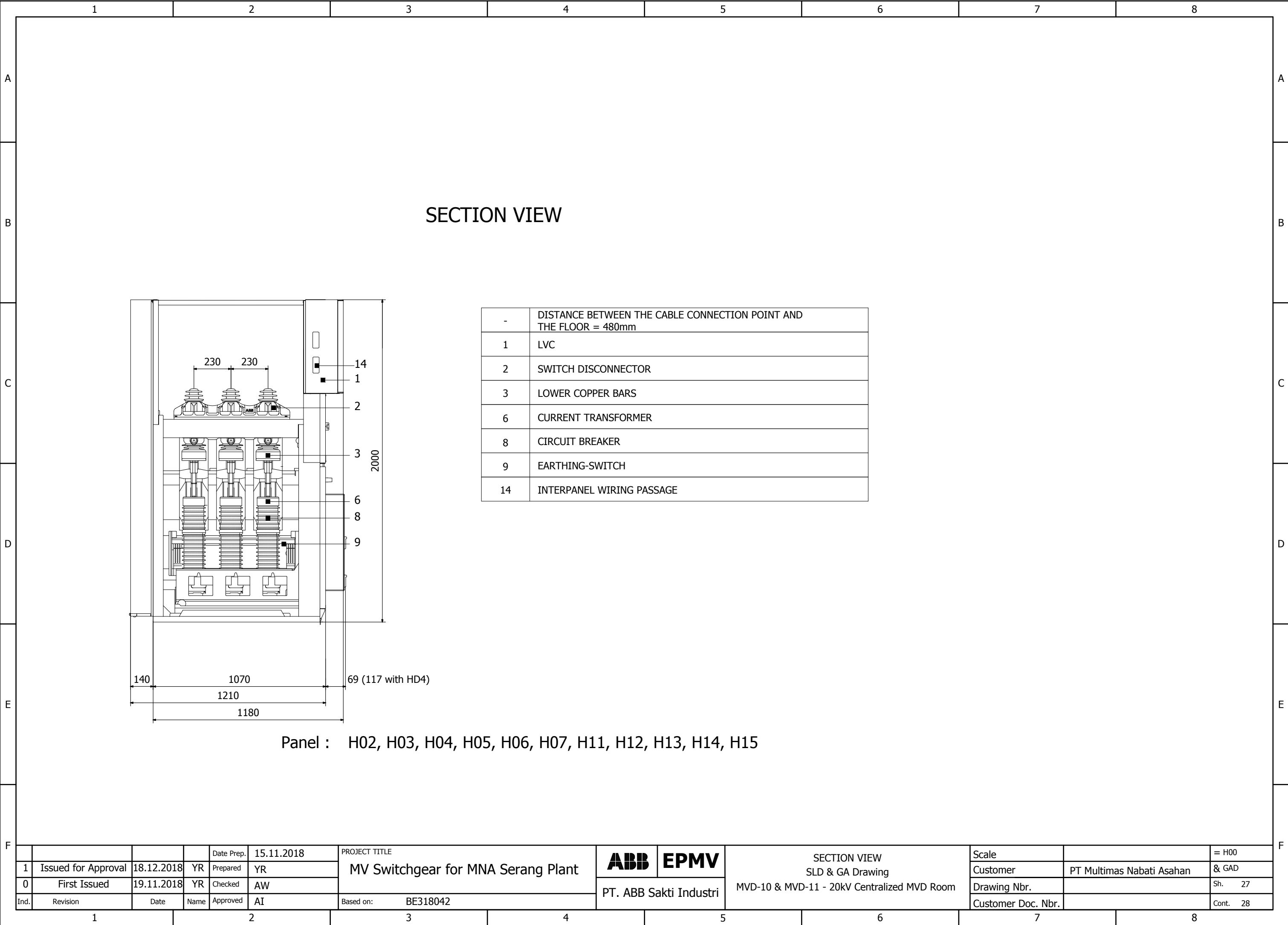


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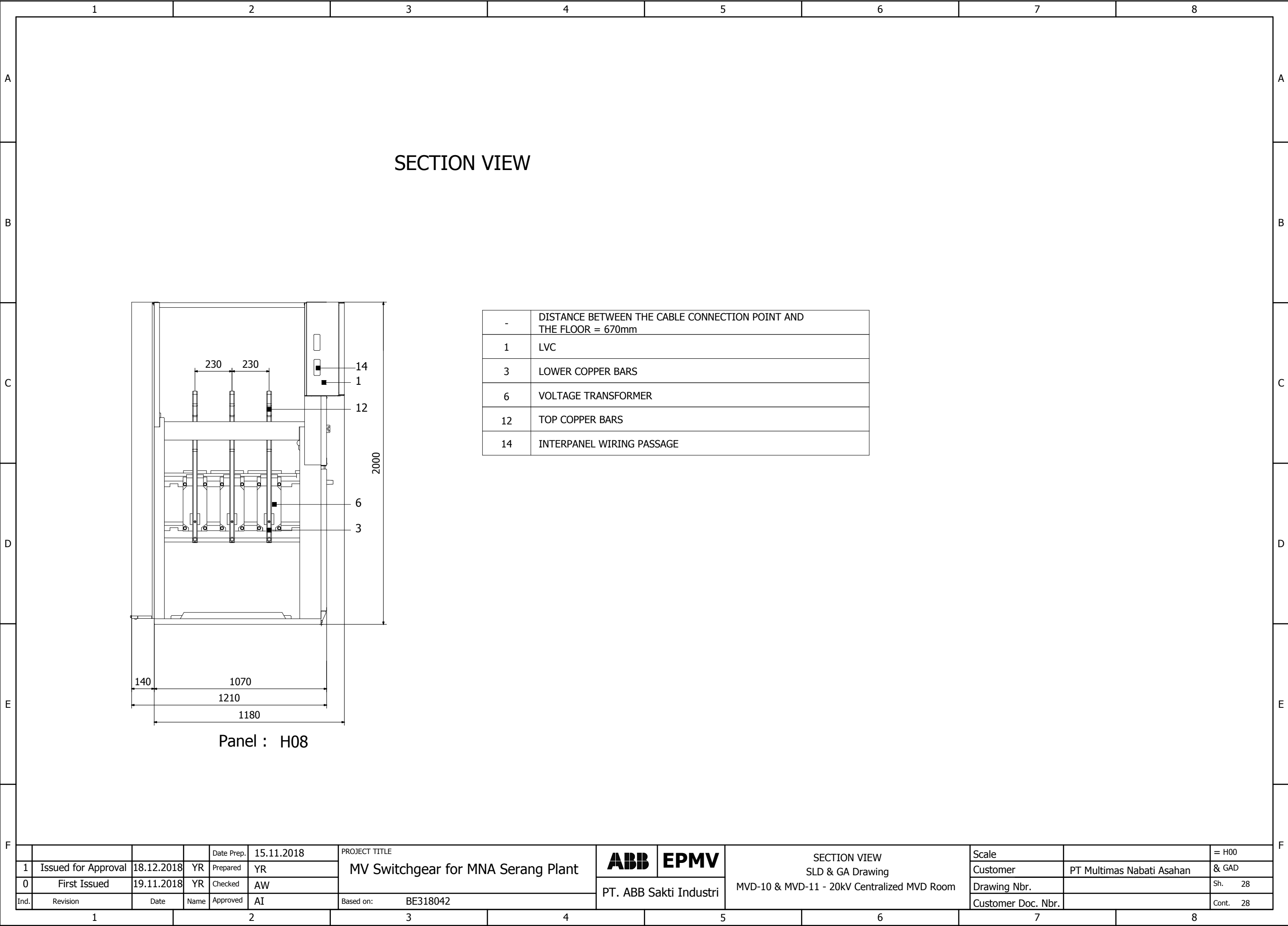


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00	
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh.	26	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.				Cont.	28

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-10 & MVD-11 - 20kV Centralized MVD Room	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 28		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28